

Version
5.0



SiftIQ

LOAD ONCE. ANALYZE EVERYWHERE.

USER GUIDE

The Complete Guide to Data Analysis,
Visualization, Automation, and Reporting



ANALYZE



VISUALIZE



SECURE



COMPARE



REPORT



AUTOMATE



CONNECT



MANAGE



POWERFUL
Advanced tools for
every type of data
analysis.



INTELLIGENT
AI-powered insights
and natural language
exploration.



EFFICIENT
Streamline your workflow
and get more done in
less time.



SECURE
Protect sensitive data
with enterprise-grade
security.



SiftIQ

Data Intelligence
Made Simple.



USER GUIDE
Version 5.0

SIFTIQ User Guide

Complete Reference Manual

Version: 1.0

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Product: SiftIQ Desktop Data Analysis Application

Audience: End-Users & Power Users

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Preface

This manual is the complete reference guide for SiftIQ, a desktop data analysis application that unifies spreadsheet-style data browsing, advanced filtering, one-click charting, AI-assisted querying, report building, and secure data workflows in a single Windows desktop environment.

All twenty-four chapters cover every major module available in SiftIQ v1.0. Whether you are a first-time user or an experienced data professional, this guide provides step-by-step instructions, worked examples, and reference tables for every feature.

Intended audience: End-users and power users, including business analysts, data professionals, IT staff, and anyone responsible for loading, exploring, and reporting on structured data.

Accessing this guide: You can open this guide from within SiftIQ at Help → User Guide.

Conventions used in this guide:

- Menu paths are written as Menu → Item.
- Button labels appear in bold, for example, **Apply Filters**.
- Keyboard shortcuts appear in parentheses, for example, Ctrl+C.
- Note boxes provide important information to be aware of.
- Tip boxes offer suggestions for more efficient use.
- File paths and code appear in a monospace font.

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Chapter 1 — Getting Started

SiftIQ is a desktop data analysis application that brings together the most common data workflows into one integrated environment. It is designed to run entirely on your Windows PC without requiring a browser, server configuration, or cloud subscription for core functionality.

SiftIQ combines the following capabilities:

- Spreadsheet-style browsing of your data
- Fast filtering and saved filters
- One-click charting (Visualization panel)
- A chat assistant that can: answer questions about your data, create charts and tables, apply filters, create derived columns, and run deterministic "AI Actions" (safe, pre-approved analysis templates)

Workflow Studio adds a captured timeline of your work so you can inspect step details, replay supported actions, export workflow history, and remove individual steps when needed. Workflow Studio also includes a SQLite-backed workflow history library so you can save workflows, recover autosaved workflow progress, and pull earlier workflow timelines back into the current studio without changing the active file.

1.1 Where Things Live

Understanding SiftIQ's layout will help you navigate efficiently. The interface is divided into several clearly defined areas:

Area	Contents
Top menu and command bar	File / Edit / Data / View / Help menus, plus the autosave toggle, Save button, Browse button, current file/path display, search box, search/filter button, and input-record button
Left sidebar	Quick navigation for Home, Dashboard, Report, Workflow, Compare, Clean, Security, Charts, Notes, SQL, SFTP, Chat, and Settings
Main workspace	Your current sheet table, Home, Dashboard, Report Builder, Workflow Studio, and maximized SQL view
Lower workspace panel	Filter panel and chart/visualization area (when those controls are active)
Right-side workspace panels	Chat, Visualization, Security, Comparison, Clean, SQL, Notes, and Integrations/Settings
Full workspace	Home, Dashboard, Report Builder, Workflow Studio, Tasks, and SFTP

Area	Contents
panels	

Figure 1.1 – The SiftIQ Home screen showing the left sidebar with all navigation icons, the main workspace with recent files, and the Tasks panel on the right.

1.2 Themes and Appearance

Open **Settings** from the left sidebar to change the app theme. SiftIQ ships with four built-in themes to suit different working environments and personal preferences.

Available themes:

- Light
- Dark
- Blue
- Excel

Themes affect: main app surfaces and side panels; dropdowns, buttons, and icons; Home, Clean, Charts, and Data Protection panels; and Report Builder styling so report editing matches the active app look.

 **Note** Your selected theme is saved and restored the next time you open SiftIQ. The Excel theme uses light surfaces with Excel-style green accents. If a panel is already open when you switch themes, SiftIQ refreshes that panel to match the new theme.

1.3 Show or Hide Left Sidebar Icons

You can customize which tools appear in the left sidebar to keep your workspace focused on the features you use most frequently.

Figure 1.2 – The SiftIQ main interface showing the left sidebar navigation icons and the data table in the main workspace, with a student dataset loaded.

How to customize sidebar visibility:

1. Right-click any left sidebar icon, or right-click an empty area in the left sidebar.

2. Use the visibility menu to check or uncheck icons such as **Home, Clean, Compare, Security, Charts, Notes, Dashboard, Report, Workflow, Chat, SQL, SFTP**, and **Settings**.
3. Use **Show All Icons** to restore the full sidebar at any time.

 **Note** Hiding an icon removes it from the left sidebar only; it does not disable the underlying feature. Sidebar visibility choices are saved and restored when you reopen the app.

1.4 Internal Service and Diagnostics

SiftIQ v5 starts a local background service when the desktop app opens. The service runs entirely on your own computer and supports faster data operations, SQL-style file queries, and future web/mobile interfaces. Normal desktop use does not require you to manage this service.

What to expect:

- SiftIQ starts the service automatically when needed.
- If the service is already running, SiftIQ reuses it.
- If the service is unavailable, desktop features can fall back to local processing where supported.
- You should not need to open a browser, enter a server address, or answer technical questions to use the desktop app.

Diagnostic file locations:

- **Startup/service snapshot:** %LOCALAPPDATA%\SiftIQ-QTFluencyWidget
v3\runtime_diagnostics.json
- **Desktop runtime snapshot:** Stored in the SiftIQ local state database under the runtime /
diagnostics entry

Chapter 1 introduced SiftIQ as a Windows desktop data analysis application and provided an orientation to its interface layout: the top command bar, the left sidebar navigation, the main workspace, and the side panels. You learned how to change themes, customize left sidebar visibility, and understand the background service that powers faster data operations. With these foundations in place, you are ready to begin loading data into SiftIQ.

Chapter 2 — Loading Data (Files + URLs)

Before you can filter, chart, or analyze data in SiftIQ, you must load a dataset. SiftIQ supports loading data from local files on your computer or from publicly accessible URLs, including Google Sheets and hosted Excel files.

2.1 Open a Local File

Menu: File → Open...

Supported file types:

Format	Extension(s)	Notes
CSV	.csv	Comma-separated values; most common format
Excel	.xlsx / .xls	Both modern and legacy Excel formats supported
JSON / JSONL	.json / .jsonl	Normalized into sheet-friendly tables automatically

After loading, Sheet1 is created or updated with your dataset. Filters and charts operate on the currently active Sheet.

 **Tip** For very large CSV files, SiftIQ loads them efficiently using its background service. If you notice a longer-than-expected load time, check the status bar at the bottom of the interface — it will display progress and row counts such as **Original: 49,987 | Filtered: 49,987** once loading is complete.

2.2 Load from a URL (Google Sheet / Excel Link)

Menu: [View](#) → [Integrations](#)

In the Integrations panel:

4. Use "**Load Google Sheet or Excel by URL**"
5. Paste the URL into the provided field
6. Click **Load**

 **Tip** If the URL requires authentication, use a share link that your computer can access directly (for example, an “Anyone with the link can view” Google Sheets share link). If loading fails, check Details & Logs for the specific error message to help diagnose the issue.

Chapter 2 covered SiftIQ's two primary data-loading paths: opening local files (CSV, Excel, JSON/JSONL) via [File](#) → [Open...](#), and loading remote data from Google Sheets or Excel URLs via the Integrations panel. Once data is loaded, the active sheet table is populated and ready for filtering, visualization, and analysis.

Chapter 3 — Sheets (Tabs) + Session Workflow

SiftIQ supports multiple "Sheets" (tabs) within a session, enabling you to keep your original dataset intact, create new Sheets from chat results, and compare datasets side-by-side using the Comparison panel.

3.1 "Show Full File" Toggle

In the status bar you may see **Show Full File**:

- **On:** The table view shows the full dataset, regardless of active filters
- **Off:** The table view shows only the currently filtered dataset (recommended for analysis)

3.2 Save/Load a Session

Menu: File → Save Session... / File → Load Session...

Use sessions to preserve:

- Sheets/tabs
- Filters and filtered views
- Charts and dashboard tiles (when available)
- Sheet column display styles (when available)
- Workflow history captured in Workflow Studio

3.3 Sheet Column Display Styles

Sheet tables support per-column display styling for quick visual analysis without modifying the underlying data.

Available modes:

Mode	Description
Inline Chart	Horizontal in-cell bar based on the column range
Comparison Bar	Centered red/green bar for positive vs negative numeric values
Trend Line	Compact sparkline-style trend beside the value
Pill	Heatmap-style pill with centered text
Column Color	Consistent accent color applied to the entire column

How to apply a column display style:

7. Right-click a column in the sheet table
8. Open **Column Display**
9. Choose a mode or color

 **Note** These styles are visual only and do not change the underlying data. Sheet column display styles persist when you reopen the same file or sheet.

3.4 Copying and Pasting Sheet Data

Sheet tables support direct clipboard copy/paste for quick data transfer between SiftIQ and other desktop applications.

Action	Shortcut	Behavior
Copy selection	Ctrl+C	Copies selected cell ranges as tab-delimited rows/columns; full row selections include the row header plus column headers; full column selections include the column header plus column values
Paste data	Ctrl+V	Pastes tabular clipboard data starting at the current cell

 **Note** Pasted values update the active sheet data, not just the visible grid.

3.5 Workflow Studio

Menu: [View → Workflow Studio](#)

Workflow Studio is the built-in activity timeline for the current sheet. It automatically captures replayable actions and informational steps, including:

- Loading data
- Applying filters
- Generating charts
- Opening workflow starters
- Saving sessions
- Dashboard changes
- Comparison actions
- Data protection changes

Figure 3.1 – Workflow Studio showing the captured step timeline on the left, the step detail panel in the center, and the Dashboard Panel on the right. Workflow action buttons (Replay, Save, Export, etc.) are visible at the bottom.

What you can do in Workflow Studio:

- Review what happened in chronological order
- Inspect structured details for the selected step
- Replay supported actions
- Copy or export a single step
- Delete a single step from the captured history
- Export or reset the full workflow

3.5.1 Workflow Studio Layout

Area	Description
Workflow Steps	Timeline of captured steps displayed on the left side
Step Details	Structured metadata, payload, and raw log for the selected step
Step Actions	Replay Action, Copy Step, Export Step, Delete Step
Workflow actions bar	Replay Workflow, Save Workflow, Save to History, Pull History, Import Workflow, Export Workflow, Reset Workflow

3.5.2 Workflow History

Workflow History is the built-in library of saved and autosaved workflow timelines, backed by a local SQLite database.

Action	Description
Save to History	Creates a manual saved workflow entry from the current timeline
Autosave	Workflow progress is autosaved in the background as a rolling history entry
Pull History	Opens an in-app popup where you can browse, preview, and pull a saved or autosaved workflow into the current Workflow Studio

 **Important** Pulling workflow history does **not** load or switch the original dataset, and does **not** replace the active file. It only restores the selected workflow steps into the current Workflow Studio. If Workflow Studio already contains steps, SiftIQ asks before replacing them.

What you see in the history popup: workflow title, saved timestamp, whether the entry is Autosave or Manual save, dataset name, source path, sheet name, summary, and captured step list.

3.5.3 Workflow Starters

Workflow Studio includes quick starters for common analysis flows:

- **Secure Data**
- **Build Dashboard**
- **Explore Charts**
- **Export Recipe**

AutoGPT Analyst

These starters open the relevant workspace and record the action in the workflow timeline automatically.

3.5.4 AutoGPT + Multistep Workflow Processing

SiftIQ includes AutoGPT-style workflow composition for multistep processing. These features are implemented inside SiftIQ and use the app's existing panels, API client, replay system, and Workflow Studio timeline.

Available AutoGPT workflow capabilities:

- AutoGPT Analyst starter: opens the ML/AutoGPT path and can seed replayable Train Model and Payment Risk Analysis steps when a target column is selected.
- Individual workflow steps: add ML Train, ML Predict, AutoGPT Risk, or an Any SiftIQ Action step from Workflow Studio.
- AutoGPT sequence: compose a Train -> Predict -> Payment Risk Analysis sequence as a set of replayable workflow events.
- Goal composer: describe a goal such as running a SQL query to a new sheet, uploading results to SFTP, preparing ML/AutoGPT parameters, or routing the goal back through Chat for action handling.
- Native replay handling: when replay sends a chat prompt, SiftIQ first tries native handlers for sheet splitting, PII masking, SFTP upload, bin columns, comparison columns, clean commands, and column styling before falling back to normal Chat.

- Progress behavior: Workflow Studio actions that are started outside Chat may still use the bottom/status area. Chat-triggered workflow actions report progress through Chat messages and the Chat progress indicator.

 **Note** Exported workflow step files use the .workflow-step.json format. Exported workflow files use the .workflow.json format. Replay is only available for steps marked as replayable.

Chapter 3 covered SiftIQ's multi-sheet session model, including saving and loading sessions, the Show Full File toggle, per-column display styles, and clipboard copy/paste. The chapter concluded with a detailed look at Workflow Studio — SiftIQ's built-in activity timeline — including its layout, step-level actions, the Workflow History library with autosave support, and quick-start workflow templates.

Chapter 4 — Filtering (Filter Panel + Search)

Filtering is one of the fastest ways to narrow your dataset before asking questions, creating charts, or building reports. SiftIQ's Filter panel offers a full set of comparison operators, support for saved filters, and a global text search for quick lookups.

4.1 Filter Panel Basics

The Filter panel includes:

- **Saved Filters** dropdown plus **Save**, **Rename**, and **Delete** controls
- **Filter Controls** list showing your current filter rules
- Buttons: **Add Filter**, **Apply Filters**, **Reset Filters**

4.2 How to Create a Filter

10. Click **Add Filter**
11. Choose: Column, Operator (equals, contains, >, <, etc.), Value
12. Click **Apply Filters**

After applying, the status bar updates to show the filtered result count, for example: **Original: 49,987 |**
Filtered: 12,340

4.3 Common Operators

Operator	Description	Best Used For
equals (=)	Exact match	Codes like 2017, F, Y
contains	Substring match	"Full-Time", "Nursing"
not contains	Excludes rows containing the text	Excluding specific categories
startswith	Prefix match	Values beginning with a code or prefix
endswith	Suffix match	Values ending with a code or suffix
regex	Regular expression pattern	Advanced pattern matching
!=	Not equal	Excluding a single value
> / >= / < / <=	Numeric comparisons	Years, counts, scores, GPA
in	Value is in a list — example: in [CA, FL, NY]	Multiple specific values
not in	Value is not in a list	Excluding multiple values
between	Inside a numeric range — example: GPA between 3.0 and 4.0	Ranges and windows
is null	Missing/empty value check	Finding blank cells
is not null	Non-empty value check	Confirming data presence

4.3.1 Regex Operator — Examples

The **regex** operator accepts standard regular expression patterns. The table below provides ready-to-use examples:

Pattern	What It Matches
<code>^[^\@\\s]+\@[^\@\\s]+\.[^\@\\s]+\$</code>	Emails (simple validation)
<code>^\(?\d{3}\)?[-.\s]?\d{3}[-.\s]?\d{4}\$</code>	US-style phone numbers
<code>^Pri</code>	Starts with "Pri"
<code>LLC\$</code>	Ends with "LLC"
<code>^(Primary Secondary)\$</code>	Exactly "Primary" or "Secondary"
<code>^S-\d{5}\$</code>	Student ID format "S-12345"
<code>^[A-Z]{2}\d{4}\$</code>	Two letters + four digits (e.g., AB1234)
<code>\b(19 20)\d{2}\b</code>	Any 4-digit year (1900–2099)
<code>\bhold\b</code>	Whole word match for "hold"

Using regex in Chat:

- Filter: AccountHolderType regex `^Pri`
- Filter: AccountHolderType regex `(Primary|Secondary)`

- Filter: Email regex `^[^\s@]+\@[^\s]+\.[^\s]+$`

 **Note** Regex is typically case-sensitive. For case-insensitive matches, prefix your pattern with `(?i)` — for example: `(?i)^pri`.

4.4 Save a Filter

13. Create and apply your desired filters
14. Click **Save** in the Filter panel
15. Enter a descriptive name for the filter set

4.5 Load a Saved Filter

16. Pick it from the **Saved Filters** dropdown
17. SiftIQ loads and applies it automatically

4.6 Reset Filters

Menu: `Edit → Reset Filters` or click **Reset Filters** in the Filter panel.

This removes all active filter rules and returns the table view to the full (unfiltered) dataset.

4.7 Search Filter (Global Text Search)

The **Search** box in the top command bar searches across all columns simultaneously as text, returning rows where **any** column contains the search term.

Tool	Best For
Search box	Quick "find anything" lookups — e.g., locating a specific name, ID, or keyword anywhere in the dataset
Filter panel	Precise, multi-column, multi-condition rules with operators, saved as reusable filter sets

Chapter 4 covered the full filtering capability in SiftIQ, from basic Add Filter / Apply Filters / Reset Filters operations to the complete list of supported operators including the advanced regex operator. You also learned how to save and load named filter sets for reuse, and how the global Search box differs from precise Filter panel rules.

Chapter 5 — Visualization Panel (Charts/Graphs)

Menu: View → Visualization

The Visualization panel lets you build charts directly from your data without using Chat. It gives you explicit control over axes, aggregation methods, and chart type through a structured builder interface.

Figure 5.1 – The Visualization Builder panel showing X-Axis, Y-Axis, Series, Aggregation, and chart type options (Bar, Bar Horizontal, Bar Stacked, Bar Grouped, Line, Area, Pie, Donut, Nested Pie, Scatter), along with Generate Chart and Add to Dashboard buttons.

5.1 Chart Inputs

Input	Description
X	Category or time axis — the dimension being charted
Y	Numeric measure, or select "count of rows" to count records
Series (optional)	Splits the chart by another categorical column (e.g., gender, region)
Aggregation	Sum / Average / Count / Min / Max
Use count of rows	Count records instead of using a numeric Y column

5.2 Chart Types Available

- Bar
- Bar (Horizontal)
- Bar (Stacked)
- Bar (Grouped)
- Line

- Area
- Pie
- Donut
- Nested Pie
- Scatter

5.3 Create a Chart

18. Choose X, Y (and optionally Series)
19. Choose an Aggregation method
20. Choose a chart type from the available options
21. Click **Generate Chart**

 **Tip** Charts operate on the currently filtered dataset. Apply your filters before opening the Visualization panel to ensure your chart reflects the intended data slice. The status bar shows **Original: 49,987 | Filtered: 49,987** to confirm the active row count.

5.4 Add to Dashboard (from Visualization Panel)

After generating a chart, two buttons appear:

- **Add to Dashboard** — adds the chart to your Dashboard silently in the background
- **Open Dashboard** — adds the chart and immediately opens the Dashboard overlay

Chapter 5 introduced the Visualization panel, SiftIQ's structured chart builder. You learned about the four chart inputs (X, Y, Series, Aggregation), the ten available chart types ranging from Bar and Line to Scatter and Nested Pie, and how to generate and save charts to the Dashboard. The Visualization panel complements Chat-driven chart creation by offering explicit, structured control over every chart parameter.

Chapter 6 — Dashboard (Saved Tiles)

The Dashboard is SiftIQ's visual summary workspace where you can save charts and tables as persistent tiles and add lightweight KPI cards to track key metrics. Unlike a reporting view, the Dashboard is interactive and tile-based, letting you arrange and resize content freely.

Figure 6.1 – The SiftIQ interface with the Dashboard Panel open on the right side, showing a horizontal bar chart (Ethnicity vs ParentIncome) and a stacked bar chart (Department vs AttendanceRate) alongside the main data table.

Typical Dashboard workflow:

22. Create a chart from the Visualization panel or Chat
23. Click **Add to Dashboard**
24. Click **Open Dashboard** to view and arrange tiles

 **Tip** Dashboard tiles often use the filtered dataset at the time of creation. When you open the Dashboard, the Saved Filters panel auto-collapses to stay compact. Expand it when you want to edit filters. Dashboard table tiles keep the full available result by default and use scrolling for additional rows and columns.

6.1 Working with Tiles (Drag / Resize)

Inside the Dashboard overlay you can rearrange and resize tiles to create a layout that works for your analysis:

- **Drag** a tile by grabbing its header bar and moving it to a new position
- **Resize** a tile using the resize handle located at the bottom-right corner
- Tiles **snap to a grid** and will not overlap — other tiles push around each other automatically

6.2 KPI Cards (Metric Tiles)

KPI cards are compact tiles that display a single computed number, ideal for at-a-glance executive metrics.

To add a KPI card:

25. Open the Dashboard
26. Click **KPI** (this opens a dropdown menu)
27. Choose a metric type
28. For column-based metrics, pick the target column from the submenu

Supported KPI metrics:

Metric	Description
Row Count	Total number of rows in the current dataset/filter
Unique Count	Distinct value count for a chosen column
Sum	Total sum of a chosen numeric column
Average	Mean value of a chosen numeric column
Percentage	Percent of rows where the chosen column is not null ("percent filled")

 **Note** KPI cards follow your current dashboard scope/filters the same way chart tiles do.

Figure 6.2 – The full Dashboard view showing KPI summary tiles (Row Count: 49,987; Unique Student: 49,987; Avg Balance: 2,500; % VisaType Filled: 9.81%), multiple charts (horizontal bar, stacked bar, area, pie), and detail tables for Unique StudentID and Avg Balance by AdmissionYear.

6.3 Table Tiles

The **Table** button (next to **KPI**) opens a dropdown for adding table-type tiles to the Dashboard.

To add a preview table tile:

29. Open the Dashboard
30. Click **Table**
31. Click **Preview**

To add a calculated table tile:

32. Open the Dashboard
33. Click **Table → Calculated Table**

34. (Optional) Enter a **Title** for the tile

35. Choose one or more **Group by** columns (you can search the column list)

36. Choose a **Value** aggregation: Sum / Average / Min / Max / Count (non-null) / Rows

37. (Optional) Set **Label**, **Sort descending**, and/or **Limit**

38. Click **Add**

 **Note** Calculated tables are built from your current filtered dataset. The tile may briefly show “Calculating...” while it computes, particularly for large datasets.

6.4 Dashboard Table Styling

Dashboard table tiles support the same column display styling system available in sheet tables:

- **Inline Chart**
- **Comparison Bar**
- **Trend Line**
- **Pill**
- **Column colors**

Dashboard table styling is saved with the tile and is replayable from Workflow Studio.

Chapter 6 covered the Dashboard — SiftIQ's tile-based summary workspace. You learned how to add chart tiles, KPI cards, and table tiles; how to drag, resize, and arrange tiles on the Dashboard grid; how to use the five supported KPI metric types; and how to create calculated table tiles with grouping and aggregation. Dashboard table styling mirrors the sheet column display styling system for consistency.

Chapter 7 — Chat (Ask Questions, Build Charts, Apply Filters)

Menu: [View → Chat](#)

Chat is SiftIQ's natural-language interface for interacting with your data. It is designed as a "do what I mean" assistant that can execute a wide range of data tasks without requiring you to use menus or form-based controls. Chat supports:

- Answering questions about your dataset
- Creating charts and tables
- Applying filters from natural language
- Creating derived columns
- Creating comparison columns
- Styling sheet columns
- Saving and sending active data to SFTP
- Running AI Actions

Running AutoGPT-style multistep workflow commands

7.1 Where Chat Results Appear

Chat can produce results in three formats:

- Plain text explanation
- An inline table
- An inline chart

Buttons that may appear under a result:

- **Add to Dashboard**
-  **Add Table to Sheet**
- **Download Response**

7.2 Copying Chat Text

Assistant and user message text can be selected directly in the chat transcript for copy/paste.

Suggestion rows still behave like clickable actions — copying text does not change normal chat behavior.

7.3 Chat Sample Privacy

Settings includes a **Send sample data** checkbox for Chat. Leave it unchecked when you want Chat to use schema and summary context only.

When **Send sample data** is checked, SiftIQ provides the following privacy options:

Option	Description
Masked sample	Sends sample context with sensitive-looking values masked
Dummy sample	Sends generated fake values that preserve the shape of the data
Schema only	Sends column/schema context without sample rows or top values
Full sample	Sends real sample rows and values — use only when the data is safe to share

7.4 Ask a Question (Examples)

- "How many students are in this dataset?"
- "Show top 10 majors by count."
- "Average GPA by EnrollmentStatus."
- "Create a bar chart of count by AdmissionYear."

7.5 Apply Filters from Chat

You can phrase a filter request in plain English and Chat will populate the Filter Controls accordingly:

- "Show students where AdmissionYear = 2024 and EnrollmentStatus = Full-Time"
- "Filter to GradYear <= 2026"
- "Only include rows where Major contains Nursing"

7.6 Add to Dashboard via Text Command

In Chat, type: `add to dashboard` — if a chart is currently available, SiftIQ adds it to your Dashboard automatically.

7.7 Save and Send to SFTP from Chat

Chat can save the active sheet data to a local export file and upload that file to the currently connected SFTP server.

Example prompts:

- save and sftp this file
- upload this dataset to SFTP
- send current sheet to SFTP
- save to SFTP as payroll.csv
- save to SFTP as payroll.xlsx

What happens: SiftIQ uses the active sheet/current filtered data → saves a local export file in the SFTP panel's local workspace → uploads that file to the current remote SFTP folder → the SFTP Transfer Queue and Activity Log show the upload result.

 **Note** You must connect to SFTP first. If SFTP is not connected, Chat opens the SFTP panel and prompts you to connect before the upload can proceed.

7.8 "Natural Language Query..." Dialog

Menu: Edit → Natural Language Query... — Shortcut: **Ctrl+K**

Use this dialog when you want quick, filter-like questions without switching to the Chat panel. Especially useful in **Natural Language** provider mode (see Chapter 17).

7.9 Style Sheet Columns from Chat

Chat can apply visual styling directly to the active sheet column using plain-language commands:

- make Points inline chart
- make Balance comparison bar
- set Score to trend line
- set Status to pill
- color Outcome blue
- clear style on Age

 **Note** These commands target the active sheet and apply the same styling path as the column right-click menu. Matching is case-insensitive and supports simple fuzzy column-name matching.

7.10 Create Comparison Columns from Chat

Chat can create a new comparison column from two existing columns and automatically apply **Comparison Bar** styling.

Example: create a comparison on credits and debits column

What happens: SiftIQ creates a new derived column using **left - right**, adds it to calculated-column definitions, and styles it as **Comparison Bar**. The comparison is created in the background so the UI stays responsive on larger sheets.

7.11 Recreate or Update a Template/Sample File Uploaded in Chat

Chat can use a template or sample file that you upload/attach in Chat to recreate or update an extract from the active sheet.

Supported attachment types:

- Saved SiftIQ extract template files in `.json` or `.csv`
- JSON sample files that represent the structure you want to recreate

Typical flow:

39. Open the sheet that contains the source data
40. Upload/attach a template or JSON sample file in Chat
41. Review the inline mapping panel that appears in chat
42. Adjust field mappings if needed
43. Click **Recreate File**
44. Save or download the generated result

45. Optionally send the saved output to SFTP

7.11.1 Inline Template Mapping Panel

When a recreatable template is attached, Chat displays an inline mapping panel listing template fields and their current matches from the active sheet. The panel supports:

- Suggested matches from active-sheet columns
- Per-field dropdown selection
- Mapped/unmapped status counts
- **Show Unmapped** to focus on unresolved fields
- **Apply Mappings**
- **Recreate File**

You can also manage mappings by typing directly in Chat:

- `show template fields`
- `show unresolved template fields`
- `map Student.StudentID to ID_Number`
- `use EmailAddress for Student.Contact.Email`
- `unmap Student.Contact.Phone`
- `apply template mappings`

After Chat generates a recreated template response, use **Download Response** to save the full output.

Fenced or raw JSON responses save as `.json`; other response types save as `.txt`.

7.12 AutoGPT + Multistep Chat Commands

Chat can route several multistep commands directly through SiftIQ's workflow and AutoGPT-style action handlers. These commands run inside the Chat experience: the Chat progress indicator turns on while the command is executing, and completion or failure is reported in the transcript instead of the bottom status bar.

Supported command families:

- Split sheets by a column, optionally combined with PII masking and SFTP upload.
- Mask or redact likely PII fields using the Security/Data Protection replay path.
- Save and upload the active sheet or current result through the SFTP panel.
- Create workflow builder steps or send workflow commands for replay.
- Run ML commands from Chat: ML train, ML predict, and ML drift.
- Run AutoGPT payment-risk analysis against a selected target column.

Example prompts:

- split into one sheet per Region and upload to SFTP
- create a sheet for each Department and mask PII
- ML train target=Outcome model=Outcomelikelihood framework=baseline
- ML predict modelid=1
- ML drift modelid=1
- AutoGPT payment risk target=Status threshold=0.4 top=20 model=Statuslikelihood

What happens: SiftIQ records eligible actions in Workflow Studio, uses replayable payloads where supported, and keeps chat-driven progress inside the Chat panel. Non-chat workflows can still use bottom/status messages for workspace feedback.

Chapter 7 covered SiftIQ's Chat panel — the natural language interface for answering data questions, building charts, applying filters, creating and styling columns, managing SFTP uploads, and recreating template files with inline field mapping. Chat result types (text, table, chart) each carry action buttons for adding to the Dashboard, adding to a Sheet, or downloading. Privacy controls for sample data give you granular control over what context is sent to the AI provider.

Chapter 8 — Percentage / Rate Questions

SiftIQ can answer "percent of rows matching X" style questions efficiently through Chat. This chapter shows how to phrase percentage and rate questions for best results.

8.1 Single Condition Percentages

For straightforward percentage questions with one condition:

- "What percent of students have EnrollmentStatus = Full-Time?"
- "Percentage of Gender is F"
- "What percent are from FL?"

8.2 Percentages with Multiple Conditions (Recommended)

Phrase your question with explicit column names and values for multi-condition percentages:

- "What percent of students have AdmissionYear = 2024 and EnrollmentStatus = Full-Time?"
- "What percent have GradYear <= 2026 and AdmissionYear = 2017?"

 **Tip** Always use explicit column names when combining multiple conditions. Ambiguous phrasing (for example, "from 2024 full time") may be interpreted differently. Using the exact column name (for example, AdmissionYear, EnrollmentStatus) makes the query unambiguous.

8.3 Graduation Rate (Cohort Within a Year)

Graduation rate computations require careful phrasing to define both the denominator (the cohort) and the numerator (those who graduated).

Component	Definition	Example
Denominator	The cohort (students admitted in that year)	Students with AdmissionYear = 2017
Numerator	Cohort members who graduated	Those same students where GradYear <= 2026

Example prompt: "Graduation rate for AdmissionYear = 2017 where GradYear <= 2026"

For a stricter four-year graduation rate definition:

"Graduation rate for AdmissionYear = 2017 where GradYear <= 2021"

Chapter 8 demonstrated how to phrase percentage and rate questions in SiftIQ Chat, from simple single-condition percentages to multi-condition combinations and cohort-based graduation rate calculations. The key best practice is to always use explicit column names and values in multi-condition questions for reliable results.

Chapter 9 — Derived Columns (Create New Columns)

Derived columns let you add computed fields to your dataset without modifying the original data source. All derived columns are created safely through Chat natural language commands — no code writing required.

9.1 When to Create a Derived Column

Use derived columns to:

- Create flags (e.g., Graduated = True/False)
- Compute differences (GradYear - AdmissionYear)
- Compute date differences (days between two date columns)
- Group/bin columns (age ranges, top categories)
- Create comparison columns from two existing numeric columns

9.2 Derived Column Examples

Goal	Example Prompt
Boolean flag	"Create a column Graduated where GradYear <= 2026"
Numeric duration	"Create a column YearsToGraduate = GradYear - AdmissionYear"
Comparison column	"Create a comparison on Credits and Debits column" — creates a new derived column using Credits - Debits
Date difference	"Create a column DaysBetween = days between EnrollmentDate and GraduationDate"

9.3 Conditional Derived Columns (IF / CASE)

Conditional derived columns evaluate a condition and return different values based on whether the condition is true or false. The THEN/ELSE result can be a **literal value**, a **column reference**, or a simple expression.

9.3.1 IF / THEN / ELSE Syntax

```
Create<New Column>= if<condition>then<result>else<result>
```

Examples:

- Create a column Transaction Amount = if TransactionType == "Credit" then Amount else Amount * -1
- Create a column RiskLabel = if Score >= 700 then "Low" else "High"

 **Note** For equality, you can use = or == — SiftIQ treats them the same. Always put quotes around text or string values in expressions.

9.3.2 CASE WHEN Syntax

```
Create<New Column>= case when<condition>then<result>else<result>end
```

Example:

```
Create a column Transaction Amount = case when TransactionType = "Credit" then Amount else  
Amount * -1 end
```

9.4 Advanced: Multi-Condition Flags (AND / OR)

- "Create a column IsActiveFullTime where EnrollmentStatus = Full-Time AND Status = Active"
- "Create a column IsRecentGrad where GradYear >= 2022 OR GradYear is null"

 **Tip** If results look wrong, re-phrase with explicit column names and the words AND or OR spelled out in full. Vague phrasing can sometimes lead to incorrect column resolution. Performance note: comparison-column creation is computed in the background so the UI stays responsive on larger sheets.

Industry Use Cases — Derived Columns

SaaS — Unit Economics Calculation

A SaaS product analyst creates a "Monthly Recurring Revenue per Seat" derived column by dividing the ARR column by active seat count. The metric is immediately filterable — the analyst surfaces all accounts where MRR per seat declined more than 10% quarter-over-quarter, feeding the expansion revenue team's target list for the week.

Telecommunications — Network Performance Metrics

A network operations engineer derives a "Packet Loss %" column from raw sent-packet and received-packet fields across a 2-million-row network log export. The derived column is immediately chartable — plotting packet loss by node identifies three routers exceeding the 0.5% SLA threshold before the morning NOC briefing begins.

Agriculture — Yield Efficiency Analysis

A farm operations manager creates a "Yield per Acre" column by dividing total harvest weight by field acreage across 200 field records. The derived column feeds a chart comparing yield efficiency by crop variety and irrigation method, informing the following season's planting and water allocation plan — no data engineer required.

Chapter 9 covered derived column creation in SiftIQ — from simple boolean flags and arithmetic columns to date-difference columns, IF/THEN/ELSE conditionals, CASE WHEN expressions, and multi-condition AND/OR flags. All derived columns are created through Chat natural language without writing code, and comparison columns are automatically styled as Comparison Bar for immediate visual analysis.

Chapter 10 — AI Actions (Deterministic Tools)

AI Actions are **pre-approved, deterministic** analysis tools built into SiftIQ. They are designed to be repeatable, safe (no arbitrary code execution), and fast — making them ideal for data validation, outlier detection, and structured analysis tasks that need to produce consistent results every time.

10.1 What Each AI Action Is For

AI Action	Purpose
Validate schema	"What columns do I have and what looks messy/missing?" — checks types, missing values, and samples
Find duplicates	"Do I have duplicated records?" — detects duplicates by selected columns
Detect outliers	"Which numeric values look unusual?" — uses z-score method
Suggest analyses	"What should I chart next?" — data-driven visualization ideas; no LLM required
Derive column	"Create a new calculated column" — date diffs, arithmetic, flags (safe operations only)
Bin column	"Group a column into buckets" — age ranges, top-k categories
Rate	"Percent of rows that match one or more conditions (optionally within a cohort)"

 **Tip** If you want transparency into the analysis logic, include “show code” in your question to ask the assistant to include the generated logic when supported by the current action.

How to use AI Actions: In Chat, ask in plain English — SiftIQ selects the best action automatically. You can also follow **Suggestions** (🌟 Auto-Suggest, covered in Chapter 11) which use the Suggest analyses action to recommend relevant visualizations.

Industry Use Cases — AI Actions

Compliance — Regulatory Submission Preparation

A compliance analyst runs SiftIQ's deduplication AI Action on a 90,000-row vendor master record before submitting it to a financial regulator. The action identifies 1,400 duplicate entries — including name-variant duplicates that manual VLOOKUP approaches had missed — and produces a clean, submission-ready file in under three minutes.

Financial Audit — Multi-Source Date Normalization

An audit team merges 12 source files from different accounting systems, each using a different date format. SiftIQ's date normalization AI Action standardizes all date fields in a single deterministic operation, eliminating the manual transformation step that previously consumed half a day per engagement.

Healthcare Quality — Anomalous Lab Result Detection

A data steward at a hospital network runs SiftIQ's outlier detection AI Action on a laboratory results dataset. The action flags 340 records with values outside clinically defined reference ranges, tagging them for physician review before the dataset loads into the analytics warehouse — preventing downstream quality metric distortion.

Chapter 10 introduced AI Actions — SiftIQ's library of deterministic, pre-approved analysis tools. These tools cover schema validation, duplicate detection, outlier detection, analysis suggestions, derived column creation, column binning, and rate computation. Unlike general LLM responses, AI Actions produce consistent, repeatable results and do not execute arbitrary code.

Chapter 11 — Suggestions (✨ Auto-Suggest)

The Chat panel includes a **Suggestions** button labeled with ✨. This feature provides automatically generated, data-aware visualization recommendations based on the structure and content of your current dataset.

How it works:

46. Click the ✨ (Suggestions) button in the Chat panel
47. SiftIQ analyzes the current dataset — its column types, cardinality, and data distributions
48. A short list of recommended visualizations is displayed
49. Click any suggestion to generate that chart immediately

This feature is designed to be:

- **Local and fast** — no LLM call required
- **Privacy-friendly** — analysis runs on your desktop, not in the cloud
- **A starting point** — generated charts can be further customized via the Visualization panel

 **Tip** Suggestions work best after you have applied filters to focus on a specific subset of data. The recommendations will reflect the filtered dataset, helping you find patterns within the segment you care about most.

Industry Use Cases — Auto-Suggest

Market Research — Survey Analysis Acceleration

A market research analyst loads a 5,000-row NPS survey export. SiftIQ's Auto-Suggest immediately recommends "Chart NPS score distribution by region" and "Filter to detractors (score 0-6) for verbatim analysis." Accepting both suggestions with one click compresses the initial exploratory analysis from 45 minutes to under 5 minutes.

Logistics — Carrier Performance Discovery

A logistics team loads a route performance dataset and SiftIQ suggests "Compare on-time delivery rate by carrier" and "Chart average delivery delay by origin region." Accepting the carrier comparison suggestion reveals that one carrier accounts for 68% of all delays — a finding that would have required a multi-step manual analysis without the suggestion engine.

Non-profit — Donor Segmentation Insight

A non-profit grants team loads a donor database and receives a suggestion to "Break down total giving by ZIP code and donor tier." Accepting it generates a geographic breakdown and tier table the development director uses to target the major gifts campaign — all without specifying which columns to analyze.

Chapter 11 covered the ✨ Suggestions feature — SiftIQ's automated visualization recommender. Suggestions analyze the current (filtered) dataset locally without requiring an LLM connection, and deliver a short menu of relevant chart ideas that can be generated with a single click.

Chapter 12 — Add Table to Sheet / Add to Dashboard

SiftIQ provides several ways to promote analysis results — chat tables, chat charts, and visualization charts — into persistent workspace items such as new Sheet tabs and Dashboard tiles.

12.1 Add a Table Result to a New Sheet

After a Chat response produces a table result:

50. Click  **Add Table to Sheet** below the Chat result

SiftIQ creates a new Sheet tab containing that result table. Use this when you want to:

- Keep intermediate results as a reference during analysis
- Chart the result using the Visualization panel
- Compare it with another dataset in the Comparison panel
- Export it as a standalone file

12.2 Add a Chart Result to the Dashboard

After generating a chart (via Chat or the Visualization panel):

51. Click **Add to Dashboard** below the result

12.3 Add a Table Tile to the Dashboard

52. Click **Open Dashboard**

53. Use **Table → Preview (Top 100)** for a quick snapshot table, or

54. Use **Table → Calculated Table** for a grouped and aggregated table tile

 **Note** Adding a table to a Sheet creates a new tab in the current session. This tab is preserved when you save the session via `File → Save Session...`. The new sheet operates identically to any other sheet — filters, visualizations, and column styling all apply.

Chapter 12 summarized the pathways for promoting Chat and Visualization results into persistent workspace items. Chat table results can be added to new Sheet tabs; chat and visualization charts can be added to the Dashboard; and Dashboard table tiles can be created as Preview or Calculated Table tiles. These workflows allow analysis results to accumulate and be reused throughout a session.

Chapter 13 — Report Builder

Menu: `View → Report Builder`

Report Builder is SiftIQ's page-layout workspace for turning your current data context into a presentation-style report. It opens in the main workspace area and hides the lower filter/chart panel while active.

You can open Report Builder by:

- Choosing `View → Report Builder`
- Clicking the **Report** side button when available
- Sending content from charts, tables, chat, or dashboard tiles into the builder

On first open, SiftIQ tries to restore an unsaved report draft. If no draft exists, it generates a starter report from the current app context. The embedded Report Builder header shows: the current dataset name, current row and column counts, and a close button to return to the prior main view.

13.1 Main Report Actions

The top report toolbar includes these primary actions:

Button	Action
Save Template	Saves the current report layout as a reusable template
Load Template	Loads a previously saved report template
Preview	Opens a page-by-page visual preview before exporting
Export	Writes the report to PDF
Auto Build Report	Automatically generates a full report from the current data context

13.2 Add Content to a Report Page

The **Elements** menu and element library provide the following content types:

Menu	Element	Description
Elements	Letterhead	Header with logo (left) and contact details (right)
Paragraph	Freeform text block	
Executive Title	Large title/heading element	
KPI	Single-metric card for report pages	
Insight Callout	Highlighted insight or key finding box	
Rectangle / Rounded	Decorative shape elements	
Rectangle / Circle		
Bar Chart / Line Chart	Embedded chart elements	
Logo/Image	Image upload element	
Table	Current Table	Embeds the current sheet table
Preview (Top 100) / Calculated Table	Preview snapshot or grouped/aggregated table	
Templates	Blank Page	Empty page starter
Executive Summary	Pre-formatted executive summary layout	
Student Analytics	Education-focused analytics layout	
Financial Snapshot	Finance-focused dashboard layout	

13.3 Letterhead Element

The **Letterhead** element creates a professional report header with your logo and contact details:

- The logo sits on the **left side** of the report header
- Contact details are aligned to the **far right** like a traditional letterhead
- The image is shown without a border box
- PDF export uses the same letterhead layout as the on-screen report

How to add a Letterhead:

55. Choose **Elements** → **Letterhead**

56. Select a logo image if you want one

57. Edit the contact details in the right-side editor

13.4 Preview and Export

- Use **Preview** to open a page-by-page visual preview before exporting to confirm layout and content
- Use **Export** to write the completed report to a PDF file

 **Note** Preview only works when the report has at least one page. If the report is empty, add at least one element or template page before using Preview.

Industry Use Cases — Report Builder

Consulting — Client Deliverable Generation

A strategy consulting firm uses SiftIQ's Report Builder to generate structured deliverables from client financial data at the end of each engagement phase. Analysts configure the report template once — executive summary, trend charts, data tables — and regenerate client-ready output from updated datasets without reformatting. Monthly report turnaround drops from two days to two hours.

Government — Legislative Budget Reporting

A municipal budget office uses the Report Builder to produce formatted annual expenditure reports for legislative review. The template includes department-level tables, variance charts, and a narrative section from the Notes panel. When budget figures are revised, the analyst refreshes the data and regenerates — formatting, headers, and structure stay intact automatically.

Real Estate — Investor Packet Generation

A property management company generates monthly rental income reports for each investor using the Report Builder. Each report pulls from a filtered view of the master property dataset, includes occupancy and revenue charts, and exports as a PDF. Generating 40 individual investor reports — previously a full day of manual formatting — now takes 25 minutes.

Chapter 13 covered Report Builder — SiftIQ's page-layout environment for creating polished, presentation-ready reports. You learned about the main toolbar actions (Save Template, Load Template, Preview, Export, Auto Build Report), the full element library including Letterhead, KPI, charts, tables, and text blocks, and the four included template layouts. Reports are exported to PDF with the same visual fidelity as the on-screen editing view.

Chapter 14 — Skills and Data Integrations

Menu: [View → Integrations](#)

SiftIQ supports three AI assistant engines ("providers") that power the Chat panel and natural language features. Each provider has different requirements, privacy characteristics, and capability levels.

Provider	Type	Requires	Best For
OpenAI	Cloud	API key + internet	Complex reasoning, chart planning, multi-step analysis
Ollama	Local or cloud	Ollama installed (local) or Ollama Cloud account	Privacy-sensitive workflows, offline use
Natural Language	Local only	Nothing — built in	Simple filtering, counting, offline use with no setup

 **Note** Integration choices (provider + model) and API keys are saved locally for your Windows user profile. If you switch computers or Windows user accounts, you will need to configure them again on the new profile.

The Integrations panel also provides:

- **Load Google Sheet or Excel by URL** (see Chapter 2.2)
- **Theme** selector — Light, Dark, Blue, Excel
- **Privacy** settings — Send sample data, Masked/Dummy/Schema/Full sample
- **Panel Role** — Custom configuration options

Industry Use Cases — Integrations (OpenAI / Ollama / NLP)

Legal Technology — Contract Clause Summarization

A legal technology firm connects OpenAI through SiftIQ's integration panel to summarize key clauses from uploaded contract datasets. Attorneys load 500-row datasets with full agreement text and ask the NLP integration, "Summarize the termination clause for each contract." Output writes as a new derived column — producing a reviewable clause summary table in minutes rather than hours of manual reading.

Life Sciences — Clinical Study Entity Extraction

A biotech research team uses SiftIQ's NLP integration with OpenAI to extract drug names, dosages, and adverse event terms from free-text clinical study export fields. Extracted entities write to new columns, enabling structured analysis across 3,000 study records that were previously unsearchable due to unstructured text content.

Retail Intelligence — Automated Product Commentary

A retail merchandising team uses Ollama locally through SiftIQ to generate product performance commentary from structured SKU data. Each row's sales velocity, margin, and return rate feeds the model, which writes a two-sentence performance summary to a new column. Category managers use these summaries in buyer meeting prep — eliminating manual write-ups for 800 SKUs per review cycle.

Chapter 14 introduced SiftIQ's three AI provider options — OpenAI (cloud), Ollama (local or cloud), and Natural Language (local, no setup) — and explained how to choose the right provider based on your privacy, offline, and capability requirements. The Integrations panel also hosts the URL data loader, theme selector, and sample data privacy settings.

Chapter 15 — Ollama Setup and Model Integration

Ollama provides a local or cloud-based model execution runtime that SiftIQ can use as its AI provider. Local Ollama runs models entirely on your Windows computer with no data leaving your machine.

15.1 First-Run Installation Prompt

On first start, SiftIQ may detect that Ollama is missing and offer to install it automatically. If **Winget** (Windows Package Manager) is available on your system, SiftIQ can install Ollama without requiring you to visit an external website.

15.2 Configure Ollama in Integrations

58. Open **View → Integrations**

59. Select **Ollama** as the provider

60. Choose runtime:

- **Local** — runs entirely on your machine
- **Cloud** — uses Ollama Cloud (requires sign-in)

61. Choose a model from the **Ollama Model** dropdown

62. Use **Refresh** to reload the model list

63. Use **Install** (local runtime) to install a model, or **Pull** (cloud runtime) to pull the selected model

 **Tip** If SiftIQ reports that Ollama is not running, open the Ollama desktop application from the Windows system tray or Start menu and wait for it to initialize, then return to SiftIQ and try again. Larger models may require several gigabytes of disk space and significant RAM — check model requirements before installing.

Industry Use Cases — Ollama Setup

Healthcare — HIPAA-Compliant Local AI Analysis

A hospital data analytics team configures Ollama locally through SiftIQ so no patient data ever leaves their on-premises infrastructure. Analysts run NLP-based ICD code extraction and clinical note summarization directly on patient datasets — capabilities that would require months of cloud procurement approval — fully compliant with HIPAA data residency requirements.

Financial Compliance — Air-Gap Anomaly Detection

A regulated bank's compliance team runs Ollama locally to analyze transaction anomaly datasets without transmitting sensitive financial records to a cloud API. The local model flags narrative transaction descriptions matching known fraud patterns, feeding the SAR filing workflow entirely within the bank's secure network perimeter.

Defense — Classified Dataset Analysis

An intelligence unit deploys SiftIQ with Ollama in an air-gapped environment to perform natural language analysis on classified operational datasets. Because Ollama runs entirely locally with no outbound network dependency, the unit achieves NLP-grade analytical capability on sensitive data without any cloud connectivity or external API exposure.

Industry Use Cases — Ollama Setup

Chapter 15 covered Ollama setup within SiftIQ, including the first-run auto-installation prompt (using Winget), configuring the Local vs. Cloud runtime, and managing models via Refresh, Install, and Pull. Ollama local runtime is the recommended choice for privacy-sensitive environments where data must not leave the desktop.

Chapter 16 — OpenAI Setup (API Keys + Models)

OpenAI provides SiftIQ's most capable AI assistance via cloud-based language models. Using OpenAI requires an API key from your OpenAI account and an active internet connection.

16.1 Add Your API Key

64. Open [View](#) → [Integrations](#)

65. Select **OpenAI** as the provider

66. Paste your key into the **OpenAI API Key** field

67. Click **Save Key**

16.2 Choose a Model

In the Integrations panel, select an **OpenAI Model** from the dropdown. Available models depend on your OpenAI account and organization settings.

 **Tip** If SiftIQ cannot connect to OpenAI: (1) verify your API key is valid and has not expired, (2) confirm your internet connection is active, and (3) check your OpenAI organization or project settings to ensure the selected model is permitted for your account tier.

Industry Use Cases — OpenAI Setup

Marketing — AI-Driven Campaign Copy Generation

A digital marketing agency configures OpenAI in SiftIQ to generate campaign copy variants from performance data. Analysts load a campaign results dataset, ask the integration to "write three subject line variants for underperforming segments," and receive model-generated copy written to new columns. Copy testing velocity increases 4x compared to the previous manual copywriting cycle.

E-Commerce — Product Description Automation

An e-commerce product team configures GPT-4 via SiftIQ's OpenAI integration to draft product descriptions from structured attribute data. Each row's color, material, dimensions, and category feed the model, which writes a complete description to a new column. Catalog content for 2,000 new SKUs is drafted in a single SiftIQ session — a task that previously required a two-week content sprint.

Media Production — Audience Research Summarization

A film production company's development team uses OpenAI through SiftIQ to summarize audience research datasets. Survey verbatim responses across 8,000 respondents compress into thematic summaries by demographic segment — giving writers and producers an AI-synthesized view of audience sentiment before greenlight decisions are made.

Industry Use Cases — OpenAI Setup

Chapter 16 covered OpenAI provider setup in SiftIQ — adding and saving an API key, and selecting a model from the Integrations panel. OpenAI provides the highest capability for complex multi-step analysis, chart planning, and nuanced natural language queries. API keys are stored locally in your Windows user profile.

Chapter 17 — Natural Language Mode (No Keys)

Natural Language mode is SiftIQ's built-in, zero-configuration AI provider. It runs entirely on your desktop computer — no API keys, no internet connection, and no model download required.

To enable Natural Language mode:

68. Open View → Integrations

69. Select **Natural Language**

This mode is best for:

- Straightforward filtering and counting
- Simple breakdowns and group-by queries
- Situations where you cannot use cloud models (air-gapped environments, policy restrictions)
- Quick queries without switching to a full LLM setup

 **Tip** For quick filter-like questions without opening the full Chat panel, use **Edit → Natural Language Query...(Ctrl+K)**. This dialog is especially efficient when working in Natural Language mode.

 **Note** Natural Language mode is designed for simplicity, not for complex multi-step reasoning or chart-planning tasks. For those use cases, configure Ollama (Chapter 15) or OpenAI (Chapter 16) as your provider.

Small Business — No-Code Data Analysis

The owner of a boutique retail chain uses SiftIQ's Natural Language Mode — with no API key configured — to query weekly sales data in plain English: "What were my top 5 products by revenue this week?" and "Which store had the lowest margin?" SiftIQ's built-in NLP processes each question and returns filtered views and charts, giving a non-technical owner direct access to their data without hiring an analyst.

Education — Academic Outcome Exploration

A university department head uses Natural Language Mode to explore student outcome datasets without any technical configuration. Questions like "Which courses have the highest failure rates?" and "Show me enrollment trends by major for the last four years" are answered immediately. The department head presents findings in the faculty meeting directly from SiftIQ — no analyst intermediary required.

Non-profit — Grant Spending Accountability

A non-profit executive director uses Natural Language Mode to explore grant expenditure data before board meetings. Without API keys or technical setup, the director asks, "How much of the education grant has been spent by program area?" and receives a breakdown table and chart. Board reporting preparation time drops from four hours to under 30 minutes.

Chapter 17 covered Natural Language mode — SiftIQ's zero-configuration local AI provider that requires no API keys, no internet access, and no model downloads. It is designed for simple filtering, counting, and breakdowns, and is the recommended starting point for users who need basic natural language queries in offline or restricted environments.

Chapter 18 — Security (Anonymize, Encrypt/Decrypt)

Menu: [View](#) → Security

SiftIQ's Security features are designed for protecting sensitive columns and sharing reports safely. The side panel is branded as **Data Protection**.

Figure 18.1 – The Data Protection panel showing Mode Selection (Encrypt/Decrypt), a searchable field list (StudentID, FirstName, LastName, Age, Gender, GPA, Major, Minor, CreditsCompleted, EnrollmentStatus), an encryption preview example (123-45-6789 → gAAAAABnEncrypted...), and action buttons (Encrypt All, Mask All, Preview Changes, Apply Security).

18.1 Anonymize & Export

Menu: File → Anonymize & Export...

Use **Anonymize & Export** when you want to export a de-identified dataset or remove/mask personally identifiable information (PII) before sharing. This creates a new export file — it does not modify the original loaded dataset.

18.2 Encrypt/Decrypt Workflow

Menu: File → Decrypt & Load File...

The Security/Data Protection panel supports the following modes and operations:

Feature	Description
Encrypt Data mode	Encrypts selected fields in the active sheet
Decrypt Data mode	Decrypts previously encrypted fields
Searchable field list	Find and select columns for protection
Recommended-only filter	Shows only columns flagged as likely to contain PII
Per-field actions	Encrypt, Mask, Hash, Skip, Decrypt (depending on mode)
Encrypt All / Mask All	Bulk-apply to all selected fields
Preview Changes	Shows the result in the active sheet before committing
Apply Security	Applies the selected security operations to the sheet

Workflow Studio captures: previewing data protection changes, and applying data protection changes.

 **Tip** Use Security features before exporting or sharing data that contains personal identifiers (PII) such as Social Security numbers, names, email addresses, or financial account numbers. Always use Preview Changes before applying changes so you can confirm the result looks correct.

Industry Use Cases — Security

Healthcare — Patient Data De-identification

A hospital research team uses SiftIQ's anonymization feature to remove patient names, date-of-birth, and medical record numbers from a clinical outcomes dataset before sharing it with an external research partner. The anonymized export complies with HIPAA Safe Harbor requirements — a process that previously required a dedicated data governance team engagement now completes in minutes within SiftIQ.

Financial Services — Audit File Encryption

A bank's internal audit team encrypts account number and social security number fields in shared audit workfiles before distributing them to external review partners. SiftIQ's field-level encryption ensures sensitive identifiers are masked in transit while numeric fields remain available for statistical validation — satisfying regulatory sharing requirements without exposing raw PII.

Human Resources — DEI Research Data Preparation

A people operations team anonymizes salary, performance rating, and demographic data before sharing it with an external DEI research firm. SiftIQ's anonymization removes direct identifiers while preserving the statistical integrity of the dataset, enabling the research firm to run equity analysis without accessing individually identifiable employee records.

Chapter 18 covered SiftIQ's Security/Data Protection features, including the Anonymize & Export workflow for creating de-identified export files, and the full Encrypt/Decrypt workflow in the Data Protection panel. Per-field actions (Encrypt, Mask, Hash, Skip, Decrypt), bulk operations, and the Preview Changes capability make it straightforward to protect sensitive data before sharing or exporting.

Chapter 19 — Data Tools (Clean Panel, Find New Records, Remove Duplicates)

SiftIQ's Data Tools panel provides guided cleanup and comparison utilities for maintaining data quality in the active sheet.

Figure 19.1 – The Clean panel showing the Auto Clean button, Scan Summary (49,987 Rows Loaded, 55 Columns Loaded, 0 Duplicates, 0 Blank Rows, 0 Empty Columns), and Cleanup action buttons (Save Clean Profile, Apply Changes, Undo Last Action).

19.1 Clean Panel

Open with: **View → Clean** or the **Clean** button in the left sidebar. If the Clean icon has been hidden from the left sidebar, open it from **View → Clean** or restore it through the sidebar right-click visibility menu.

The Clean panel is the guided workspace for quick cleanup actions on the active sheet. It includes:

Section	Contents
Auto Clean	One-click cleanup suggestions and common cleanup flows
Scan Summary	Row count, column count, duplicate count, blank row count, empty column count
Cleanup Actions	Remove blank rows, delete empty columns, trim extra spaces, standardize values
More Tools	Column-level normalization, value standardization, data type conversion
Duplicates	Highlight, Show, Remove, Show Unique, and Clear
Detected Issues	Review what SiftIQ found before applying additional actions

Typical Clean workflow:

- 70. Open **Clean** on the sheet you want to fix
- 71. Review the **Scan Summary** and **Detected Issues** sections
- 72. Run one or more cleanup actions (remove blanks, trim spaces, etc.)
- 73. Use duplicate tools to inspect, isolate, or remove repeated rows
- 74. Rescan if needed to confirm the sheet is in the desired state

 **Note** The Clear action in the Duplicates section restores the full visible sheet after a duplicate review. Cleanup changes update the current sheet data rather than opening a separate results window.

19.2 Find New Records...

Use to detect new rows when comparing datasets or incoming updates against an existing baseline. This is useful when you receive a refreshed export and need to identify what changed.

 **Tip** Find New Records works best when you have a clear unique identifier column (such as StudentID or TransactionID) to use as the match key between the two datasets.

19.3 Remove Duplicates...

Use to remove duplicate rows from the active sheet while keeping one copy of each unique record.

 **Tip** Choose columns that together define a unique record — for example, StudentID + Term for enrollment records, or TransactionID for financial data. Avoid using all columns as the composite key unless exact full-row duplicates are what you are targeting.

Chapter 19 covered SiftIQ's data quality tools: the Clean panel (Auto Clean, Scan Summary, Cleanup Actions, More Tools, Duplicates section, and Detected Issues), the Find New Records utility for detecting dataset additions, and Remove Duplicates for deduplicating the active sheet. These tools operate on the active sheet and provide a guided, non-destructive workflow with preview capabilities before committing changes.

Chapter 20 — Comparison Panel

Menu: View → Comparison

The Comparison panel helps you compare two datasets side-by-side — for example, Sheet1 vs. Sheet2, or a "before" snapshot vs. an "after" update.

Figure 20.1 – The Comparison panel showing the Data Comparison section with Primary file (student_data_50k_50col_enrollment compare.csv / Sheet1) selected, the right comparison file not yet selected, and action buttons: Find New, Left Join Add Columns, and Merge.

Typical comparisons supported:

- Row counts between datasets
- Schema differences (column presence and types)
- New or missing records
- Duplicate highlighting
- Duplicate-only views
- Unique-only views
- Merged output sheets

How to run a comparison:

75. The **left** side defaults to the current active file/data
76. Select a **right** comparison file or sheet
77. Add one or more **match key** pairs to define how rows are joined
78. Use **Find New** to create a sheet of rows found only on the left
79. Use **Merge** to create a merged output sheet
80. Use duplicate tools to highlight, display, or remove duplicate rows

Workflow Studio captures all major comparison actions including: added/removed comparison key pairs, highlighted/displayed duplicate rows, unique-row views, find new records, merges, and duplicate removal.

 **Tip** Use the Comparison panel together with Workflow Studio to create a reproducible comparison workflow. After setting up and running a comparison, save it to Workflow History so you can replay the same steps on a future data refresh.

Chapter 20 covered the Comparison panel — SiftIQ's side-by-side dataset comparison tool. It supports match-key-based record comparison, Find New for isolating added rows, Merge for producing combined output sheets, and duplicate row tools for reviewing and removing repeated records. All comparison actions are captured in Workflow Studio for replay and audit.

Chapter 21 — SFTP Workspace

Open with: **SFTP** in the left sidebar.

The SFTP Workspace is a two-pane file transfer view for moving files between your Windows computer and an SFTP (Secure File Transfer Protocol) server. It provides a full connection manager, file browser, and transfer queue integrated directly into SiftIQ.

What the SFTP Workspace includes:

- **Connection selector** for saved SFTP profiles
- **New Connection** for host, port, username, password, or SSH key setup
- **Saved Connections** for editing or deleting saved connection profiles
- **Remote SFTP** browser for navigating server folders and files
- **Local Workspace** browser for local files on your computer
- Actions: **Download**, **Upload to SFTP**, **Preview**, **Import to SiftIQ**, **New Folder**
- **Transfer Queue** showing upload/download progress and status
- **Activity Log** showing connection, listing, transfer, and error messages

21.1 Create or Edit a Connection

81. Open the **SFTP** workspace from the left sidebar

82. Click **New Connection**

83. Enter the connection details:

- Connection name
- Host (server address)
- Port (default: 22)
- Username
- Authentication method: **Password** or **SSH Key**
- Starting remote path (optional)

84. Click **Save** or **Save & Connect**

 **Note** Passwords and key passphrases are session-only and are not saved to disk for security reasons. Use **Saved Connections** to edit or delete a saved profile's other settings (host, port, username, path).

21.2 Browse Remote and Local Files

After connecting:

- The **Remote SFTP** pane lists the resolved server folder and its contents
- The **Local Workspace** pane lists the selected local folder on your computer
- Double-click folders to enter them
- Use **Up** or the path box to navigate to a parent folder
- Use the search boxes to filter the visible file list by name

21.3 Transfer Files

Action	How to Use	Result
Download	Select a remote file → click Download	Copies the file to the current local workspace folder
Upload to SFTP	Select a local file → click Upload to SFTP	Copies the file to the current remote SFTP folder
Import to SiftIQ	Select a remote file → click Import to SiftIQ	Downloads the file and opens it directly in SiftIQ
New Folder	Click New Folder in either pane	Creates a new folder at the current local or remote location

 **Note** Uploads go to the current remote folder shown in the **Remote SFTP** path box. Downloads go to the current local folder shown in the **Local Workspace** path box. Transfer results appear in both the **Transfer Queue** and **Activity Log**.

21.4 SFTP Troubleshooting

Issue	Resolution
Remote pane is empty after connecting	Confirm the remote path is correct; check the Activity Log for the folder being listed
Login succeeds but expected folder is missing	Enter the absolute remote path in the path box, such as /home/username
SSH key authentication fails	Confirm the private key file path is correct and the passphrase (if any) is entered correctly for this session
Upload/download stalls or fails	Check the Activity Log for the specific error; verify network connectivity to the SFTP host

Chapter 21 covered the SFTP Workspace — SiftIQ's integrated two-pane file transfer tool. You learned how to create and manage SFTP connections, browse remote and local file systems, transfer files in both directions, import remote files directly into SiftIQ, and diagnose common SFTP connectivity issues. SFTP transfers triggered from Chat (Chapter 7.7) use the same connection and workspace configured here.

Part II — Use Cases, Workflow Enhancement, and Excel Productivity

The following section extends the core product guide with practical operating patterns for teams that use SiftIQ alongside Excel. It is intended to help analysts, operations teams, finance users, delivery managers, and data-driven leaders translate product features into repeatable business workflows.

Executive Summary

SiftIQ can materially improve productivity when teams spend too much time on repetitive data preparation, validation, reconciliation, exception handling, and reporting work. Many organizations do not struggle because they lack data, but because they lack a repeatable operating flow between raw data, trusted data, and decision-ready output. SiftIQ helps create that flow.

- Ingest and normalize files from multiple sources.
- Standardize transformations that are currently done manually.
- Apply validation logic consistently.
- Isolate exceptions instead of reworking entire files.
- Track changes and produce auditable outputs.
- Feed curated datasets into Excel for fast business analysis.

When combined with Excel, SiftIQ enables a high-speed model: SiftIQ handles the heavy-lift data operations, while Excel supports interpretation, exploration, and presentation. This separation of concerns helps operations remain deterministic and auditable while keeping analysts productive in a familiar environment.

Core Productivity Model

- Acquire data.
- Standardize structure.
- Validate quality.
- Transform for purpose.
- Separate exceptions.

Chapter 22 — Troubleshooting

This chapter covers the most common issues encountered in SiftIQ and their recommended resolutions.

Issue / Message	Likely Cause	Resolution
"No data loaded."	No file has been opened in the current session	Load a file via File → Open... or load from URL via View → Integrations
"No data matches your query."	Filters are too restrictive, or the value/column name is misspelled	Broaden your filters (try fewer conditions). Double-check spelling and exact category values. Check the status bar for the current filtered row count.
OpenAI connection error	Invalid or expired API key; no internet connectivity; model not permitted	Ensure your API key is saved in View → Integrations. Verify internet connectivity. Check your OpenAI organization/project settings for model access.
Ollama error / "Ollama not running"	Ollama is not installed or not currently running	Open the Ollama desktop application and wait for it to start. Then try Integrations → Ollama → Refresh . If needed, try a different model.
Chart does not show expected data	Chart was created before applying filters, or the wrong aggregation was selected	Apply the desired filters first, then regenerate the chart. Check the X / Y / Aggregation settings in the Visualization panel.
Derived column shows wrong values	Ambiguous column name or condition phrasing	Re-phrase with explicit column names and AND/OR spelled out. Check for typos in column names.
SFTP upload/download fails	Authentication issue, wrong path, or network problem	Check the SFTP Activity Log for the specific error. Re-enter credentials for the session. Verify the remote path exists.
Left sidebar icon is missing	Icon was hidden via the sidebar visibility menu	Right-click the left sidebar → use the visibility menu to re-enable the icon, or click Show All Icons .

22.1 Collecting Diagnostic Information

If you are unable to resolve an issue using the steps above:

85. Open View → Details and Logs

86. Copy the latest errors from the log

87. Check the diagnostic files at: %LOCALAPPDATA%\SiftIQ-QTFluencyWidget

v3\runtime_diagnostics.json

 **Tip** When reporting an issue, include the contents of the log, the file type and approximate size of the dataset you were working with, the AI provider you were using, and the steps that led to the problem. This significantly speeds up diagnosis.

Chapter 22 Summary Chapter 22 provided a reference troubleshooting table for the most common SiftIQ issues including data loading errors, filter and query mismatches, OpenAI and Ollama connectivity problems, chart and derived column issues, SFTP failures, and hidden sidebar icons. The chapter concluded with instructions for collecting diagnostic logs from the Details and Logs view and local diagnostic files.

Chapter 23 — Prompt Cookbook (Copy/Paste Examples)

This chapter is a ready-to-use reference of Chat prompts and command phrases organized by task type.

Copy any example directly into the SiftIQ Chat panel.

23.1 Chat Command Phrases (Bypass / Force Behavior)

These command prefixes instruct SiftIQ Chat to treat the prompt in a specific way, overriding default planning behavior.

Command	Syntax / Example	Notes
Web search	Search for <question> Search: <question> Web: <question> Example: "Search for how many students are at the University of Miami"	Bypasses dataset execution plans. Works even if the Internet toggle is OFF. Uses the currently selected AI provider (View → Integrations)
Force Filter Controls	Filter: <conditions> Filters: <conditions> Example: "Filter: State = TN and Amount > 1000"	Populates Filter Controls and applies the filters. Requires a dataset to be loaded.
User Guide help	Help: <topic> User guide: <topic> Guide: <topic> Examples: Help: regex User guide: filters Guide: derived columns	Searches the local SiftIQ_USER_GUIDE.md and returns matching snippets. Does not require a dataset to be loaded.
Save and	save and sftp this file upload this	Requires an active dataset and an active SFTP

Command	Syntax / Example	Notes
send to SFTP	dataset to SFTP send current sheet to SFTP save to SFTP as payroll.csv	connection.
Force a chat table	Table: <question> or start with "Create a table ..." Example: "Table: count by AdmissionYear"	Nudges the planner to produce a table result instead of a text answer.
Build an agentic dashboard	Dashboard 12 Dashboard 12 replace Dashboard 12 replace with ID, Admission, Year, GPA, Dorm Dashboard 10 with AccountID, Region, Amount about revenue	Builds 10–15 tiles automatically. "replace" clears existing tiles first. Guardrails: year-like fields are dimensions; count tiles prefer Unique <EntityID>; sum used for additive fields; averages for rates/scores

23.2 Sheet Column Styling

- make Points inline chart
- make Balance comparison bar
- set Score to trend line
- set Status to pill
- color Outcome green
- clear style on Age

23.3 Create a Comparison Column

- create a comparison on credits and debits column
- add a comparison between revenue and cost
- make a comparison between credits and debits called net movement

 **Note** Creates a new derived column using left - right. The new column is automatically styled as Comparison Bar.

23.4 Filtering

- "Show students where AdmissionYear = 2024"
- "Filter to EnrollmentStatus = Full-Time"
- "Only include rows where Major contains Nursing"

- "Show rows where GradYear <= 2026 and AdmissionYear = 2017"

23.5 Counts and Breakdowns

- "How many students are there?"
- "Count by AdmissionYear"
- "Top 10 majors by count"

23.6 Percentages / Rates

- "What percent of students have EnrollmentStatus = Full-Time?"
- "What percent have AdmissionYear = 2024 and EnrollmentStatus = Full-Time?"
- "Graduation rate for AdmissionYear = 2017 where GradYear <= 2026"

23.7 Charts

- "Create a bar chart of count by AdmissionYear"
- "Create a pie chart of EnrollmentStatus"
- "Create a line chart of Average GPA by AdmissionYear"
- "Scatter plot of Credits vs GPA"

23.8 Derived Columns

- "Create a column Graduated where GradYear <= 2026"
- "Create YearsToGraduate = GradYear - AdmissionYear"
- "Create DaysBetween = days between EnrollmentDate and GraduationDate"
- Create a column Transaction Amount = if TransactionType == "Credit" then Amount else Amount * -1
- Create a column Transaction Amount = case when TransactionType = "Credit" then Amount else Amount * -1 end

23.9 Derived Comparison Columns

- "Create a comparison on Credits and Debits column"
- "Make a comparison between Revenue and Cost called Margin Delta"

23.10 Suggestions

Click ✨ **Suggestions** in the Chat panel and choose a recommended visualization from the list. No prompt required — SiftIQ analyzes your dataset and generates suggestions automatically.

23.11 Add to Dashboard (Command-Style)

Type: `add to dashboard`

If a chart is currently available from the last Chat result, SiftIQ adds it to your Dashboard silently. Combine with a chart prompt for a streamlined workflow:

88. "Create a bar chart of count by AdmissionYear"

89. "add to dashboard"

Chapter 23 provided a comprehensive Prompt Cookbook — a ready-to-use reference of Chat commands, control phrases, and example prompts organized into eleven categories: command phrases (web search, force filter, help, SFTP, table, agentic dashboard), column styling, comparison columns, filtering, counts, percentages/rates, charts, derived columns, comparison columns, auto-suggestions, and add-to-dashboard commands. Keep this chapter open as a quick reference while working in SiftIQ.

Chapter 24 — FastAPI as the Service Core of SiftIQ v5

SiftIQ v5 marks a deliberate architectural shift from a desktop-first tool into a service-first platform. At the center of that shift is FastAPI, which functions as the execution backbone for data loading, cleaning, analytics, AI-assisted recommendations, reporting, and workflow orchestration. Rather than embedding all behavior directly in UI code, SiftIQ now exposes a typed HTTP interface that can be consumed by desktop shells, future web clients, and automation pipelines.

This chapter explains how FastAPI is implemented in SiftIQ v5, why it was chosen, how requests move through the system, and what this means for maintainability, migration, and scale.

24.1 Why FastAPI in SiftIQ v5

The FastAPI adoption in SiftIQ is not cosmetic. It solves concrete engineering problems that became limiting in prior desktop-centric versions. FastAPI provides: strong request and response contracts through Pydantic models; automatic API documentation for discovery and onboarding; a clean separation between interface and computation; consistent error handling across heterogeneous operations; and a stable integration point for multiple clients.

In practical terms, this means SiftIQ features no longer depend on a single UI process to be useful. The same dataset pipeline can be called from a desktop widget, an API test, or a future browser app without duplicating logic.

Note This chapter is developer-oriented. End-users benefit from FastAPI indirectly — through faster, more reliable panel responses and the ability to interact with SiftIQ from external tools.

24.2 High-Level System Architecture

At runtime, SiftIQ v5 is organized as a thin client layer over a rich service layer, with FastAPI as the orchestration boundary.

Figure 24.1 — SiftIQ v5 Runtime Architecture
Client Layer → FastAPI Router → Service Layer → Dataset Session / Legacy Bridge

Figure 24.1: SiftIQ v5 Runtime Architecture — the desktop client consumes FastAPI endpoints; routers delegate to the service layer, which manages dataset sessions and the v4/v5 compatibility bridge.

Routers define endpoint contracts and input validation. Services hold business logic and data manipulation. Dataset state is sessionized in memory. Legacy module compatibility is managed by an import bridge, not by endpoint-level hacks.

24.3 Application Assembly and Startup Model

This chapter consolidates the Skills and Data Integrations addendum into the main SiftIQ User Guide so it reads like a native part of the book. It covers how optional skills are activated, where they appear in the app, and how to safely configure and use Outlook, Oracle, and Gmail as SQL-accessible integration sources.

24.3 Application Assembly and Startup Model

The FastAPI application is created in a factory style and then bound once at module load, giving a clear composition root where middleware and routers are registered in one place. Startup paths include: standalone API process using Uvicorn on localhost port 8000; desktop-managed startup that reuses an existing server if healthy; embedded startup mode in frozen distributions; and lazy mode where API startup is deferred unless explicitly enabled.

Operational behavior is surfaced through runtime status values and diagnostics writes, allowing support workflows to quickly identify whether the API was reused, started, embedded, timed out, or unavailable.

Tip SiftIQ does not assume one deployment style. It supports developer execution, packaged desktop execution, and mixed lifecycle control — making it adaptable across environments without code changes.

24.4 CORS and Local Development Policy

SiftIQ configures CORS for local origins commonly used by front-end dev servers. This enables immediate integration with local web tooling without modifying backend code each time a port changes. It lowers friction for multi-client evolution, keeps the API ready for web migration while still desktop-focused, and standardizes trusted local origins rather than requiring ad hoc patches.

24.5 Router Topology as Domain Boundaries

SiftIQ v5 organizes endpoints into domain routers rather than a single monolithic module. This mirrors product capabilities and makes API evolution safer.

Router Domain	Responsibility
Health and Configuration	Service liveness, runtime diagnostics
Datasets	Load, preview, paginate, and sessionize data
Cleaning Operations	Normalization, deduplication, outlier handling
Actions and Module Catalog	Run deterministic AI actions, list available modules
Filters	Apply, save, and reset filter state

Router Domain	Responsibility
Profiles	Column profiles, type inference, summary stats
Charts	Chart configuration and render data
Dashboard	Tile management and layout persistence
Utilities	Cross-cutting helpers and conversion tools
Compare	Side-by-side dataset comparison
Security	Anonymization and field-level encryption helpers
SQL	Query execution, schema introspection, saved queries
Files	File metadata, SFTP-backed file operations
Reports	Report generation and template management
Workflows	Workflow capture, replay, and history
Agents	AI-backed natural language and recommendation endpoints

This decomposition means teams can evolve domain behavior independently, and API docs remain navigable for users and testers.

24.6 Request Lifecycle in SiftIQ FastAPI

A typical request follows a predictable path through the system:

HTTP entry at router endpoint → Payload validation by Pydantic models → Service-layer execution → Optional dataset session mutation → Response model serialization → Exception mapping to HTTP status via centralized handler

The consistency of this pattern across all routers is a significant maintainability strength — any developer familiar with one router can immediately work with any other.

24.7 Dataset Session Model: Stateful by Design

SiftIQ maintains dataset sessions in memory, keyed by generated dataset identifiers. Each session keeps source metadata, a baseline dataframe, and a current working view dataframe. The dual-frame model supports a practical workflow: keep original context, apply filters and transforms to the working view, and return paginated previews and summaries from the current state.

This model is ideal for interactive analytics but has implications: session data is process-local; horizontal scaling requires future session externalization; and restarting the process drops in-memory sessions unless persisted by dedicated endpoints.

Note The dual-frame approach (baseline + working view) is what allows SiftIQ to show filters, transforms, and derived columns without ever overwriting the source data. The original dataset is always preserved in the session baseline.

24.8 Contract Design with Pydantic

SiftIQ uses explicit Pydantic model classes for health, dataset load, dataset preview, action execution, filter requests, profile responses, and configuration responses. Contract-first development provides validation before business execution, cleaner client integration, better auto-generated documentation, and lower ambiguity in endpoint behavior.

Schema design in SiftIQ is not just type safety — it is product clarity. Each model defines exactly what the endpoint expects and what it promises to return, making the API self-documenting.

24.9 Error Normalization Strategy

Rather than manually crafting HTTP errors in every endpoint, SiftIQ routes exceptions through a shared mapper:

Exception Type	HTTP Status
Missing files, missing session keys	404 Not Found
Invalid input values, malformed requests	400 Bad Request
Unexpected internal failures	500 Internal Server Error

This creates a stable client contract. A desktop panel, web app, or automation script can rely on predictable status semantics regardless of which router domain produced the error.

24.10 SQL Execution Path and Guardrails

The SQL router demonstrates a pragmatic, layered execution strategy: validate query structure and enforce SELECT or CTE patterns; attempt execution through a DuckDB-compatible path; fallback to in-memory SQLite execution when needed; return structured records with bounded limits.

Additional behavior includes in-process saved query storage and schema introspection endpoints. This shows FastAPI handling both strict validation and graceful execution fallback in one domain.

Tip Use the SQL panel's schema introspection endpoint to quickly discover column names and types before writing complex queries. This is especially useful when working with large or unfamiliar datasets.

24.11 AI and Utility Integration Through Service Adapters

Agent and utility routers do not hardcode legacy logic. They call service adapters that import computational modules through a compatibility bridge. That bridge prefers v5 modules first and only falls back to v4 modules when required. This means: migration can happen incrementally; existing proven algorithms remain available during transition; and API contracts remain stable even while internals evolve.

This is one of the most important implementation patterns in SiftIQ v5 — it allows the product to improve continuously without breaking existing workflows.

24.12 Compatibility Layer as Migration Infrastructure

The compatibility import mechanism is a strategic component, not a temporary patch. It enforces v5-first module resolution, controlled fallback to v4 path, and deconfliction when stale modules were loaded from undesired locations.

In organizational terms, this enables continuous delivery during migration. Teams can modernize one domain at a time while preserving working product behavior — a pattern that significantly reduces deployment risk.

24.13 Desktop as Client, Not Monolith

The desktop API client and launcher code illustrate the intended future direction: UI is increasingly a consumer of API contracts, not the host of business logic. Desktop behavior includes health probing before rendering panels, endpoint-driven data actions for every feature, API docs launch support for developers, and runtime status messaging for diagnostics.

This creates a clean long-term path: keep UI innovation independent of backend contracts, reuse backend logic across platforms, and increase testability through HTTP-first workflows.

24.14 Testing Philosophy: API Smoke Over UI Coupling

SiftIQ's smoke tests exercise FastAPI endpoints directly through FastAPI's test client. Covered areas include: health endpoint validity; dataset load and pagination; module action runtime and execution; filter logic integration; agent and utility exposure; and dashboard-ready endpoint behavior.

This is a deliberate quality strategy. Testing the service boundary catches regressions where they matter most and reduces dependence on fragile UI automation for core correctness.

Note FastAPI's built-in TestClient (from Starlette) allows tests to run without starting a live server. This makes SiftIQ's smoke tests fast, deterministic, and safe to run in CI/CD pipelines.

24.15 Current Tradeoffs and Operational Considerations

SiftIQ v5's FastAPI layer is strong, but it is honest to document current constraints:

Constraint	Implication
Session and workflow state are in-memory	Process restarts clear active sessions
Saved SQL queries are in-process memory	Queries do not persist across restarts without export
Authentication and role-based authorization are not yet a first-class layer	All local API traffic is currently trusted
Some security helper operations are utility-grade	Not suitable as a replacement for enterprise cryptography

These are typical for a transition-phase architecture and do not diminish the value of the current implementation. They clarify roadmap priorities for the v5-to-v6 evolution cycle.

24.16 Strategic Impact for SiftIQ

FastAPI in SiftIQ v5 should be understood as the platform boundary that enables multi-client product evolution (desktop, web, and automation on one backend); incremental migration from legacy internals without breaking the product surface; faster onboarding through typed contracts and auto-generated API docs; better observability and predictable operations through structured status semantics; and stronger testability at the service layer, independent of the UI.

In short, SiftIQ v5 is no longer just a desktop application that happens to have API calls. It is an API-centric data system with a desktop client.

Conclusion: The FastAPI implementation in SiftIQ v5 is both technically coherent and strategically important. It combines typed contracts, modular routing, centralized error semantics, domain service adapters, and compatibility-driven migration into a practical architecture that can evolve safely. FastAPI is not merely the transport layer in SiftIQ — it is the structural mechanism through which SiftIQ transitions from a single-interface tool into a reusable analytics platform.

FastAPI is SiftIQ v5's execution backbone for all data, analytics, AI, and workflow operations.

- **16 domain routers** organize endpoints by capability, making the API navigable and independently evolvable.
- **Pydantic contracts** enforce request/response validation before any business logic executes.
- The **dual-frame session model** (baseline + working view) preserves source data while supporting interactive transforms.
- A **compatibility bridge** enables incremental migration from v4 to v5 modules without API contract changes.
- The **SQL router** demonstrates defensive design: strict validation with graceful execution fallback.
- Testing targets the **service boundary directly** via FastAPI's TestClient, reducing UI automation dependency.
- Current constraints (**in-memory sessions, no auth layer**) reflect a transition-phase architecture with a clear roadmap ahead.
- **SiftIQ v5 is an API-centric data platform** — the desktop UI is now one of several possible clients.

25.1 Skills (Activation + Sidecars)

25.1.1 Why skills exist

- Keep optional enterprise features isolated from core file analysis flows.
- Allow staged rollout by environment without forcing all users to enable every integration.
- Keep provider-specific settings in dedicated config files.

25.1.2 How skills are activated

SiftIQ checks skill activation in this order:

1. Explicit skill list from the environment variable **SIFTIQ_ACTIVE_SKILLS**.
2. Managed tokens registered by the app at runtime.
3. Auto-discovery marker files for supported skills.

Examples:

```
SIFTIQ_ACTIVE_SKILLS=outlook,oracle
```

```
SIFTIQ_ACTIVE_SKILLS=gmail
```

25.1.3 Navigation: where to use skills in the app

1. Open **View** → **Integrations** to configure skill connection settings.
2. Open the **SQL** panel from the left sidebar.
3. In SQL, switch **Source** to an enabled integration such as Outlook, Oracle, or Gmail.
4. Use **Test Connection** before running scripts.

Choose a result destination:

- **Preview only** for safe checks.
- **New sheet** to append results to the current workbook session.
- **New document** to replace the workspace with a new result document.

25.1.4 Updating skill configuration safely

1. Change one setting at a time.
2. Click **Save Settings** in Integrations.
3. Click **Connect** or **Test** and confirm success.
4. Run `SELECT ... LIMIT 20` before any write operation.
5. If behavior changes unexpectedly, disable the skill and revert.

25.2 Data Integrations (Outlook / Oracle / Gmail)

25.2.1 Activation summary

- **Outlook:** set `SIFTIQ_ENABLE_OUTLOOK=1` and include `outlook` in `SIFTIQ_ACTIVE_SKILLS`.
- **Oracle:** include `oracle` in `SIFTIQ_ACTIVE_SKILLS`.
- **Gmail:** set `SIFTIQ_ENABLE_GMAIL=1` and include `gmail` in `SIFTIQ_ACTIVE_SKILLS`.

If a skill is not enabled, it should not appear as an active SQL source.

25.2.2 Outlook integration

Auth: Microsoft sign-in using device flow.

Core SQL tables: messages, inbox, folders, plus folder-derived tables.

Missing destination folders can be auto-created during move or update.

Outlook navigation and update examples:

Go to **View → Integrations** and click **Connect Outlook**.

Open the SQL panel and set source to **Outlook Mail**.

Run:

```
SELECT subject, from, received
FROM inbox
LIMIT 20;
```

List folder tables:

```
SELECT sql_table, display_name, unread_item_count
FROM folders
ORDER BY display_name;
```

Move unread invoices:

```
UPDATE inbox
SET folder = "Finance Follow Up"
WHERE is_read = false
AND subject LIKE '%invoice%';
```

Delete alert emails from a folder table:

```
DELETE FROM alerts
WHERE from LIKE '%noreply%'
AND subject LIKE '%expired%';
```

25.2.3 Oracle integration

Auth/connect: Oracle endpoint plus credentials or wallet settings.

SQL execution runs against the Oracle source when Oracle is selected.

Oracle setup note: leaving Wallet blank

If connecting to a non-wallet Oracle environment, **Wallet Location** can be blank.

Config Dir can be blank unless your DBA requires a local network or admin config.

Client Lib Dir can be blank unless explicit Oracle client libraries are required.

Oracle navigation and query examples:

Go to **View → Integrations** and save Oracle settings.

Open the SQL panel and select the Oracle source.

Validate:

```
SELECT 1 AS OK FROM DUAL;
```

Preview data:

```
SELECT *  
FROM STUDENTS  
FETCH FIRST 20 ROWS ONLY;
```

25.2.4 Gmail integration

Auth: Google OAuth installed-app flow.

Core SQL tables: messages, emails, labels, plus label-derived tables.

Gmail uses labels rather than Outlook-style folders. Move behavior is implemented as label updates.

Missing destination labels can be auto-created during move or update.

Gmail navigation and update examples:

Go to **View → Integrations** and click **Connect Gmail**.

Open the SQL panel and set source to **Gmail**.

Validate:

```
SELECT subject, from, received  
FROM messages  
LIMIT 20;
```

List labels:

```
SELECT sql_table, name, messages_unread  
FROM labels  
ORDER BY name;
```

Move promo messages:

MOVE FROM inbox

TO "marketing_archive"

WHERE subject LIKE '%promo%';

Delete (trash) newsletters:

DELETE FROM messages

WHERE from LIKE '%newsletter%'

AND subject LIKE '%weekly%';

25.2.5 Integration testing checklist

1. Open the Integrations panel and save settings.
2. Click the integration-specific connect or test button.
3. In the SQL panel, choose the integration source.
4. Run a low-risk query such as `SELECT * FROM messages LIMIT 20`.
5. Confirm result destination behavior.
6. Validate one write action with a narrow WHERE clause.
7. Re-run `SELECT` to confirm expected changes.

25.2.6 Troubleshooting

Source missing in SQL panel:

Verify `SIFTIQ_ACTIVE_SKILLS` and provider-specific enable flags.

Sign-in fails:

Confirm client settings in the integration JSON config. Re-run the connect flow and complete the browser or device prompt.

Query errors:

Start with `SELECT ... LIMIT 20`, then test `DELETE` or `MOVE` using a narrow predicate.

Oracle wallet confusion:

If wallet authentication is not required, leave wallet fields blank and test simple SQL first.

Mail update safety:

Use **Preview only** first, and use restrictive WHERE clauses before move or delete operations.

Chapter 25 added the Skills and Data Integrations guidance as the new final chapter of the SiftIQ User Guide. It explains how optional skills are activated, where integrations are configured in the app, and how Outlook, Oracle, and Gmail can be safely tested and used as SQL-accessible sources with low-risk validation and troubleshooting guidance.

SiftIQ User Guide — Complete Reference Manual

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